



Ancestral Skies Collaboration

March 6th, 2026

Attention: Harry Hawkins

We represent a collaboration of optical telescopes that are built on the lands of First Nations people in Australia, Chile, Hawaii and South Africa. Phenomena associated with Near-Earth Objects (NEOs) have long been important in reinforcing cultural law, spirituality, and the moral structure of our communities. We are putting together a proposal to use our resources in searching for NEOs and need assistance in optimising how our telescopes should work together to improve our expected detection rate.

Three of our telescopes (**AU1**, **CL1**, **SA1**) are specifically designed for NEO detection while the other three (**AU2**, **CL2**, **HI1**) are general telescopes that can also be used but are less efficient. The table below shows the efficiency in detecting NEOs for each telescope.

Telescope	AU1	CL1	SA1	AU2	CL2	HI1
Efficiency (%)	99	97	99	40	43	36

Each night we can choose a number of hours on each telescope to observe either the Main Belt (20° to 30°) or the NEO Belt (-15° to 15°). We expect a base NEO discovery rate of 2.2 per hour from the Main Belt and 3.4 per hour from the NEO Belt. For example, if we used AU1 to observe the Main Belt for 2 hours and the NEO Belt for 1 hour then we would expect $0.99 \times (2 \times 2.2 + 1 \times 3.4) = 7.7$ discoveries from the 3 hours.

We have limited access to the telescopes, depending on the day of the week. The table below shows the total observing time for each of the telescopes, along with the amount of that time that the NEO Belt is visible.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Total (hours)	4.8	5.5	6.4	4.7	5.3	5.4	6.0
NEO Belt (hours)	3.3	3.8	3.9	3.4	3.6	3.8	4.7

For example, we could use AU1 on Monday to observe the Main Belt for 4.8 hours, or we could observe the Main Belt for 1.5 hours and observe the NEO Belt for 3.3 hours.

How should we allocate observing hours to each telescope for the week? Please provide us with the optimal expected number of NEO discoveries.

478.7

Communication 2

Thank you for your initial proposal. However, we have realised that the telescopes SA1 and CL2 both have fields of view large enough to observe the Main Belt and NEO Belt simultaneously. For example, on Monday SA1 can observe the Main Belt for 4.8 hours and the NEO Belt for 3.3 hours.

Taking this into account, how should we allocate observing hours to each telescope for the week? Please provide us with the optimal expected number of NEO discoveries.

561.5

Communication 3

NEOs appear as moving points of light, and can move significantly in their orbits over short periods. Delays in processing can make follow-up observations impossible if the object moves out of the predicted area. Due to the rapid motion of NEOs and the need for prompt follow-up observations, all survey images must be processed within 24 hours of collection to enable effective orbit determination and recovery. Our total processing capacity across all six telescopes is 500 GB per day.

Each telescope produces 25 GB per hour of operation. This includes SA1 and CL2, even though they are observing both regions.

Taking our data management into account, how should we allocate observing hours to each telescope for the week? Please provide us with the optimal expected number of NEO discoveries.

456.7

Communication 4

We have noted that your proposal is not making good use of AU2 or HI1. To ensure we have a genuine collaboration between our partners, we would like all telescopes to be used for at least 10 hours during the week.

Taking this into account, how should we allocate observing hours to each telescope for the week? Please provide us with the optimal expected number of NEO discoveries.

443.7

Communication 5

Our IT team have told us that we can store unprocessed data at the end of a day but only up to a day's worth of additional data (500 GB).

Taking this data storage into account, how should we allocate observing hours to each telescope for the week? Please provide us with the optimal expected number of NEO discoveries.

We look forward to reading your final report.